Mechanism of action and adverse effects of drug used for treatment of Diabetes Mellitus

Learning outcomes

By the end of this lecture, students should be able to:

- Describe treatment of Diabetes Mellitus
- Outline Mechanism of action drug used for treatment of Diabetes Mellitus
- Describe adverse effects of drug used for treatment of Diabetes Mellitus

Introduction

- **Diabetes mellitus (DM)** is a metabolic disorder characterized by hyperglycemia, glycosuria, hyperlipidemia etc.
- A widespread pathological change is a **thickening of the capillary** resulting in vascular complications like lumen narrowing, early atherosclerosis, sclerosis of glomerular capillaries, retinopathy, neuropathy, and peripheral vascular insufficiency.
- Almost 28% of people with diabetes are undiagnosed.
- Because diabetes is a relatively common condition, practicing dentists are likely to encounter it frequently.

Types of Diabetes Mellitus

Types of Diabetes:

•Type 1: Insulin-dependent
•Type 2: Non-insulin dependent
•Gestational Diabetes

Relevance to Dental Practice:

•Increased risk of periodontal disease, delayed healing, and oral infections

Symptoms of Diabetes

- General symptoms:
  - Polyuria, Polydipsia, and Polyphagia
  - Fatigue, weight loss, blurred vision
- Oral manifestations:
  - Xerostomia, gingivitis, periodontitis, burning mouth syndrome

Treatment Options

- **Lifestyle Modifications:**
  - Diet and exercise recommendations
  - Smoking cessation
- **Pharmacological Treatments:**
  - **Oral Hypoglycemics:** Metformin, glipizide, sitagliptin, semaglutide
  - **Insulin Therapy:** Types of insulin (rapid-acting, long-acting)

Diabetes and Dental Care

- **Pre-treatment Considerations:**
  - Importance of patient history and glycemic control assessment
- **Dental Management Tips:**
  - Morning appointments (ideal for controlled patients)
  - Avoiding hypoglycemia during dental procedures
  - Antibiotic prophylaxis for high-risk cases
- **Oral Care Guidelines for Diabetic Patients:**
  - Educating patients on oral hygiene practices

INSULIN

- The hypoglycemic hormone **insulin** is synthesized in the β cells of pancreatic islets.
- The most prominent action of insulin is hypoglycemia.
- It facilitates **glucose transport across the cell membrane.** It alters the activity of enzymes involved in carbohydrate, fat, and protein metabolism in the liver, muscle, and adipose tissue to lower blood glucose levels.
- Concurrently, the actions prevent the rise in free fatty acid levels, ketone body production, and protein breakdown of the diabetic state.
- Insulin lispro, insulin aspart, and insulin glulisine have a more rapid and predictable onset, peak, and shorter duration of action than regular insulin.

Types of insulin

- ***Human insulin*** produced by recombinant DNA technology.
- The human and purified animal insulin produces less allergic reaction, insulin resistance, and injection site lipodystrophy.
- Several analogs of human insulin have the same Pharmacodynamic actions but exhibit greater stability and consistency.
- ***Insulin lispro, insulin aspart,*** **and** ***insulin glulisine*** have a more rapid and predictable onset, peak, and shorter duration of action than regular insulin.

Reactions to insulin

- Hypoglycemia is the most frequent and potentially the most serious reaction to insulin.
- Hypoglycemia can occur in any diabetic following inadvertent injection of large doses, by *missing a meal (e.g. after a dental procedure)* or by performing vigorous exercise.
- Sweating, anxiety, palpitation, tremor; and those due to glucose deprivation of brain—dizziness, headache, behavioural changes, visual disturbances, hunger, fatigue, weakness, muscular incoordination and sometimes a fall in BP.
- Allergy can also occur.

Oral Antidiabetic Drugs

1. Biguanides (e.g., Metformin):

- Reduces glucose production in the liver.
- Side Effects: GI upset, lactic acidosis (rare).

2. Sulfonylureas (e.g., Glipizide):

- Stimulates insulin secretion.
- Side Effects: Hypoglycemia, weight gain.

3. DPP-4 Inhibitors (e.g., Sitagliptin):

- Increases incretin levels to enhance insulin secretion.

4. Thiazolidinediones (e.g., Pioglitazone):

- Improves insulin sensitivity.

5. SGLT-2 Inhibitors (e.g., Empagliflozin):

- Promotes glucose excretion through urine.

Symptoms of hypoglycemia

Symptoms of Mild-to-Moderate Hypoglycemia

Shakiness

Sweating

Fast or irregular heartbeat

Dizziness or lightheadedness

Hunger

Nervousness

Change in behavior or personality

Tingling or numbness of the lips or tongue

Sleepiness

Blurred vision

Loss of coordination

Headaches

Weakness

Trouble concentrating, confusion

Paleness

Irritability
Argumentative, combative

Symptoms of Severe Hypoglycemia
Unable to eat or drink Seizures or convulsions Unconsciousness

Treatment of Hypoglycemia

- If hypoglycemia is suspected, immediate treatment should be implemented.
- Check the patient’s blood glucose levels using a glucometer. Levels that are $\leq$70 mg/dL indicate hypoglycemia.
- Provide the patient **with 15-20 grams of oral carbohydrates to eat or drink**.
- If the dental patient is not awake and/or unable to eat or drink, emergency medical help should be summoned. **Injectable glucagon**, available by prescription, signals the liver to release glucose into the bloodstream and can help restore blood glucose levels to normal in emergencies. **Glucagon may be administered while waiting for help to arrive.**

Effect of diabetes on oral health-1

- Diabetes mellitus is associated with increased incidence of many dental problems like caries tooth, periodontal disease, oral candidiasis and other infections, etc.
- Diabetes and smoking are both considered risk factors for periodontitis.
- The dentist should review the patient's medical history, take vital signs, and evaluate for oral signs and symptoms of inadequately controlled diabetes, which may be common.
- Manifestations of uncontrolled diabetes can include Xerostomia, burning sensation in the mouth (which may be related to neuropathy), impaired/delayed wound healing, increased incidence and severity of infections, secondary infection with candidiasis; parotid salivary gland enlargement; gingivitis and/or periodontitis

Effect of diabetes on oral health-2

- Dental extractions and other simple procedures under local anaesthesia do not usually pose any special problems in diabetics, provided blood sugar levels are well controlled.
- If the diabetic who is being treated with insulin/oral hypoglycemic is to miss the meals after a dental procedure, care should be taken that he/she does not develop hypoglycemia.
- For more extensive dental procedures or those to be done under general anesthesia, it is advisable to monitor urine and blood sugar levels and cover the perioperative period with regular insulin injected S.C. Severe/uncontrolled diabetics may be given an I.V. infusion of regular insulin along with 5% glucose solution during the procedure.
- Prophylactic antibiotics should be given to cover dental procedures in diabetics, particularly if they are poorly controlled.

Drug treatment of Peptic Ulcers

Learning Outcomes

By the end of this lecture, students should be able to:

Describe treatment of peptic ulcer

Outline Mechanism of action and adverse effects of drug used for treatment of peptic ulcer

Outline Mechanism of action and adverse effects of Antibiotics.

Definition and Relevance

Peptic ulcers are lesions in the gastric or duodenal mucosa caused by the action of acid and pepsin.

Relevance to Dentistry:

•Impact on systemic health and oral manifestations.

•Considerations during dental procedures (e.g., stress, NSAID use, etc.).

Causes:

- Helicobacter pylori infection.
- NSAID-induced mucosal damage.
- Stress, smoking, and alcohol.

Drugs for ulcers

A) Proton Pump Inhibitors (PPIs)

•Examples: Omeprazole.

•**Mechanism**: Irreversibly inhibit H+/K+ ATPase in gastric parietal cells.

•**Effects**: Strong acid suppression.

•**Side Effects**: Headache, diarrhoea, long-term effects include B12 deficiency.

B) H2-Receptor Antagonists (H2RAs)

•Examples: Ranitidine (withdrawn in many regions), Famotidine.

•**Mechanism**: Block histamine H2 receptors, reducing acid secretion.

•**Effects**: Moderate acid suppression.

•**Side Effects**: Headache, dizziness, rare gynecomastia (cimetidine).

Drugs for Ulcers Contd’

C) Antibiotics (for H. pylori)

•Examples: Amoxicillin, Metronidazole

•**Regimen**: Triple therapy (PPI + 2 antibiotics) or Quadruple therapy (PPI + bismuth + 2 antibiotics).

•**Effects**: Eradication of H. pylori.

•**Side Effects**: Diarrhea, resistance.

D) Antacids

•Examples: Aluminum hydroxide, Magnesium hydroxide

•**Mechanism**: Neutralize stomach acid.

•**Use**: Rapid symptom relief.

•**Side Effects**: Constipation (aluminum), diarrhea (magnesium).

Drugs for Ulcers Contd':

E. Cytoprotective Agents

- **Sucralfate**: Forms a protective barrier on the ulcer.
- **Misoprostol**: Prostaglandin analogue that enhances mucus and bicarbonate secretion (used in NSAID-induced ulcers).

F. Bismuth Subsalicylate

- **Mechanism**: Protective coating and antibacterial action.
- **Effects**: Reduces H. pylori activity, and protects mucosa.

Dental Issues

H. pylori Connection: Possible link to oral cavity (e.g., dental plaque, saliva).

NSAIDs and Ulcers: Awareness of potential oral manifestations.

Drug Interactions:

PPIs may interfere with the absorption of drugs like antifungals.

Antibiotics may cause oral side effects like candidiasis.

References

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